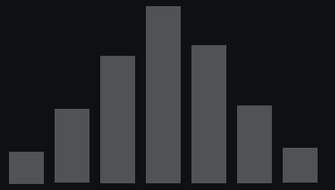
MACHINE LEARNING

DETECTING GENAI TEXT

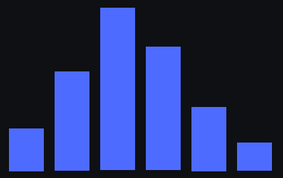
A STORY OF DISTRIBUTION SHIFT

The model was never the hard part.

TWO SAMPLES, ONE CORPUS



TRAIN · 20,000



TEST · 5,000

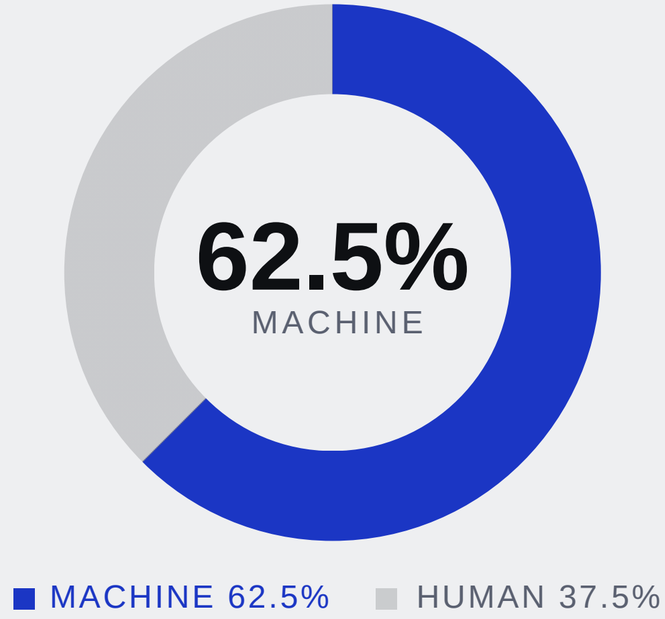
THE PROBLEM

INPUT → LABEL

Given a text: human **0** or machine **1**?

MACRO-F1 **not accuracy. Both classes weighted equally**

> “Always machine” scores 62.5% accuracy and catches zero human text.



FOUNDATIONS

PART 1

NumPy only

Logistic regression from scratch

Sigmoid with input clipping, cross-entropy loss, mini-batch gradient descent.

OURS	SKLEARN
0.____	0.____

Matched almost exactly → implementation validated.

PART 2

KNN · k = 2

PCA, then KNN



Fewer components scored *higher*: the curse of dimensionality. Schematic; exact Macro-F1 values to be filled from the report.

On the provided 5,000 opaque features, we hit a wall.

BROAD EXPLORATION

14

MODELS SCREENED

154

ENSEMBLE COMBINATIONS

SWEEP → GRID

FAMILY 01

Linear models

Linear decision boundaries over sparse features

FAMILY 02

Naïve Bayes

Word-likelihood models, fast and strong on text

FAMILY 03

Kernel & distance

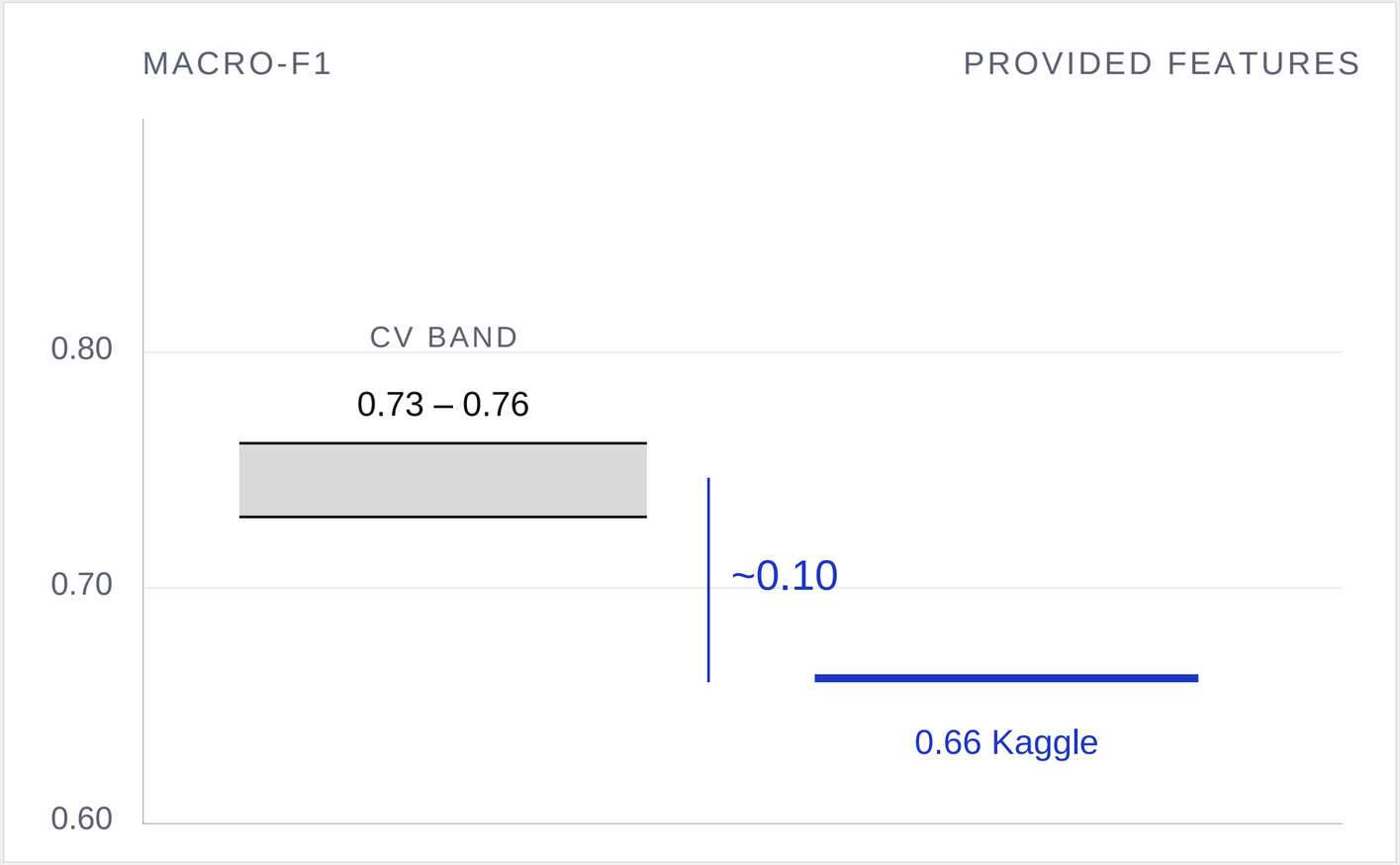
Margin and similarity methods in feature space

FAMILY 04

Tree ensembles

Many shallow decisions combined by voting

THE CEILING



CROSS-VALIDATION

0.73–0.76

LEADERBOARD

0.66

UNCLOSED GAP

~0.10

The bottleneck was the features, not the models.

FEATURES FROM RAW TEXT

VIEW 01

Word TF-IDF

Which words and word pairs appear: “in conclusion”, “as an AI”.

VIEW 02

STAR

Character n-grams

Spacing, punctuation, spelling habits. Style is topic-agnostic, so it generalises.

VIEW 03

Stylometrics

Hand-built measures: sentence length, punctuation rates.

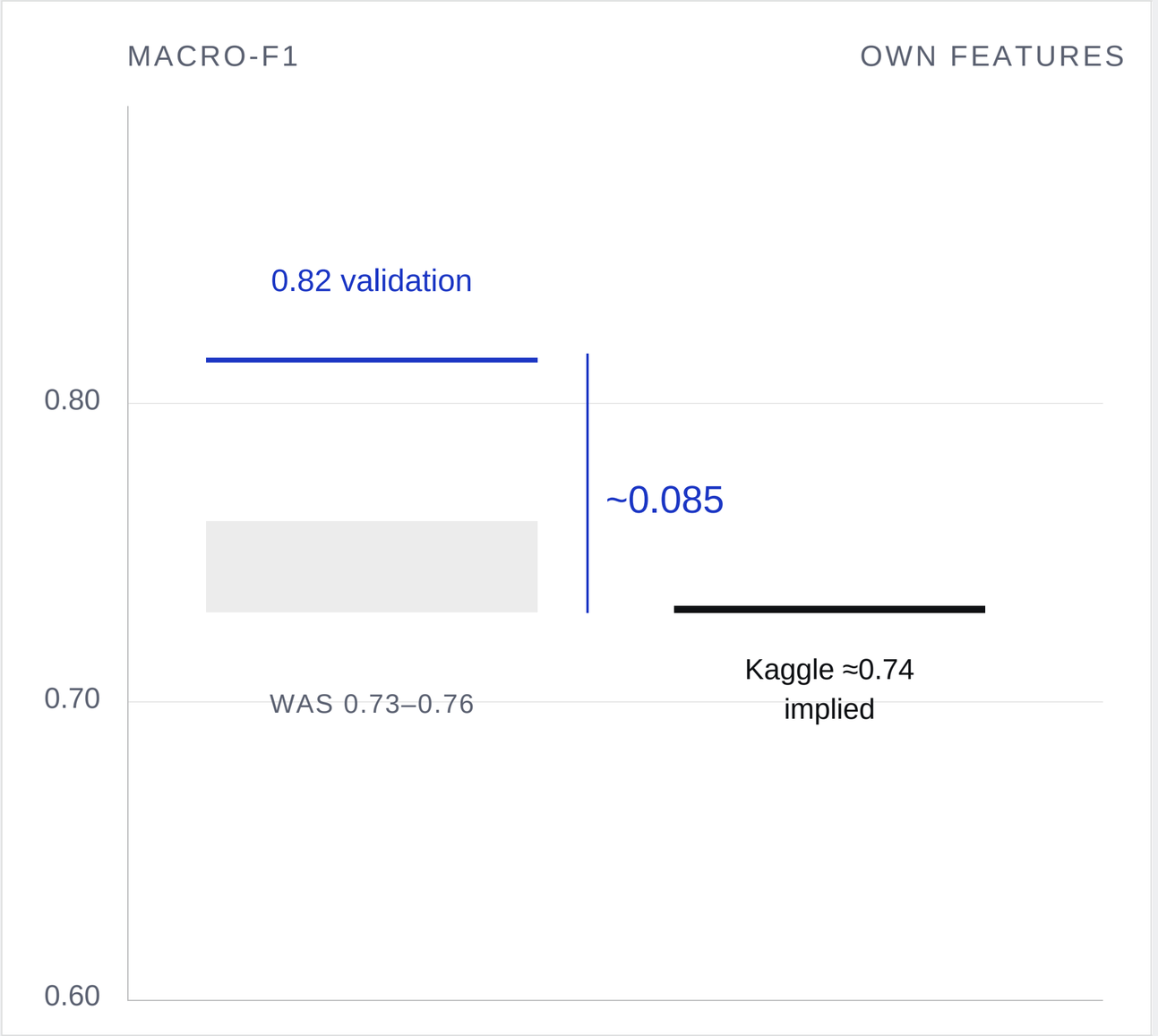
Same logistic regression, new features:

0.73 → **0.82**

THE GAP HELD

Better features lifted the whole curve…

…and the distance to the leaderboard barely moved.



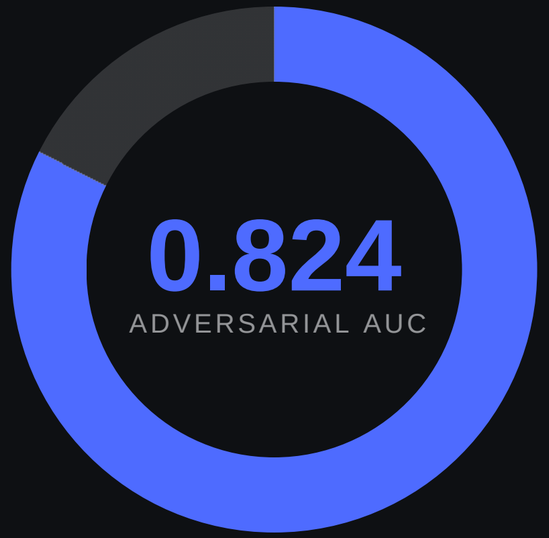
If models improve, and features improve, and the gap survives both, the gap is about neither.

DISTRIBUTION SHIFT

Train and test are not drawn from the same distribution, and a classifier trained to tell them apart proves it.

~0.085 CV overstated the leaderboard by this much

Optimising on CV was selecting the models that overfit the training set.



0.50 = INDISTINGUISHABLE · 1.00 = TRIVIALLY SEPARABLE

ONE MODEL, THREE JOBS

ONE CLASSIFIER

Adversarial model

Trained to tell a train row from a test row.

01 **Measures the shift**

AUC 0.824: how learnable the difference is

02 **Importance weights**

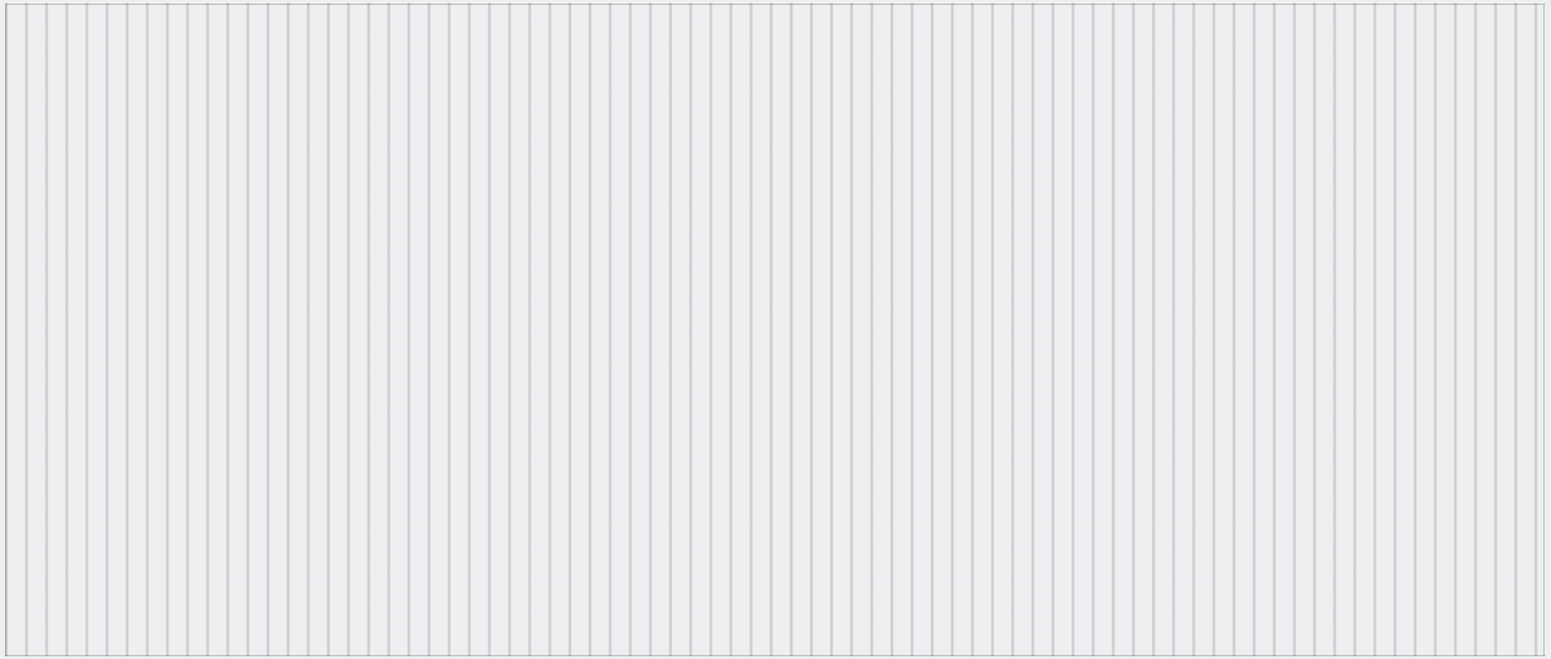
Up-weight the training rows that look test-like

03 **Shift-mimicking splits**

advQ and lenQ validation that behaves like the leaderboard

COMPRESSING AWAY THE SHIFT

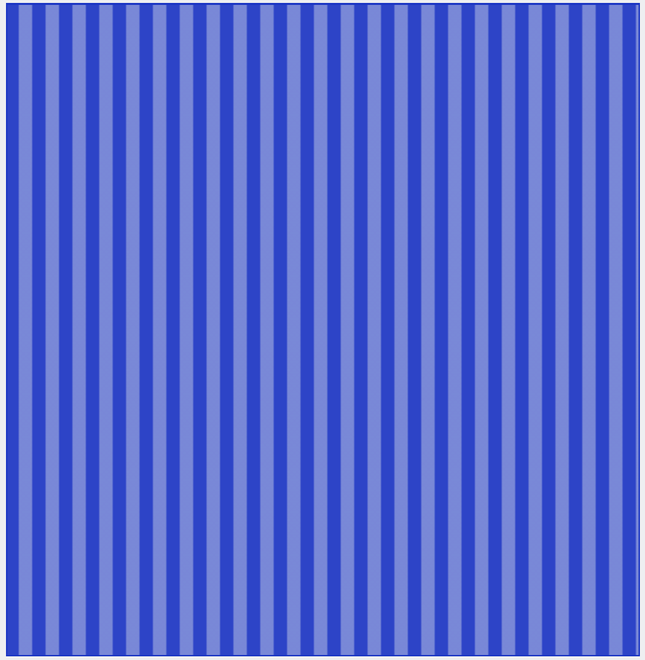
SPARSE · 180,000 COLUMNS



Trees split on rare, shift-specific tokens, losing 5–10 F1 points.

→

DENSE · TRUNCATED SVD



Loss drops to ~0.02.

SAME IDEA AS PCA **Discard the low-variance noise directions and the shift goes with them.**

FOUR REFINEMENTS

01 NBSVM

Reweight every feature by its Naïve Bayes score, then run logistic regression on top.

02 Agreement-gated self-training

Learn from unlabelled test inputs, but only trust a guess when two feature views agree.

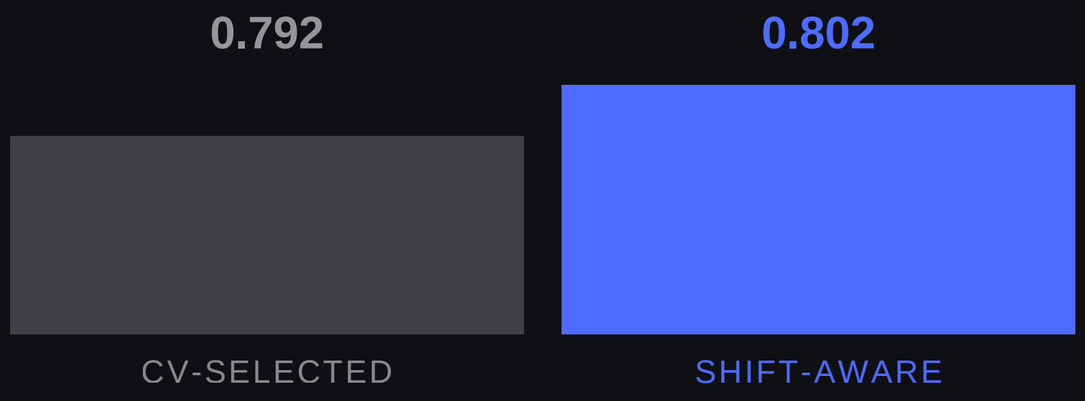
03 Near-duplicate override

~20 test inputs are near-copies of training rows → borrow the known label.

04 Calibration & threshold

Threshold moved from 0.50 to 0.59, giving a positive rate of ~0.60, estimated by leaderboard probing.

THE PAYOFF



The score crossed 0.80 exactly when we stopped trusting cross-validation.

FINAL BEST SUBMISSION · MACRO-F1

0.80242

Importance weights

advQ / lenQ splits

NBSVM + LSA

Threshold 0.59

LIMITATIONS & LESSONS

STRUCTURAL	ACCEPTED RISK	HARDEST PROBLEM	SHELF LIFE
<5%	**0.59**	**~0.085**	**1 snapshot**
of the full corpus. A small sample will always behave differently.	threshold partly fitted to the public leaderboard, not the private one.	gap that remained. Validation, not modelling, was the real work.	of machine text. Newer generators write differently.

TAKEAWAY **The win came from diagnosing the problem, not from a bigger model.**

PROBLEM → FEATURES → SHIFT → 0.802

We didn’t find a better model. We found out what our numbers were hiding.